fastqueue

A small, dependency-free job queue for Node.js with retries, priorities and rate limits.

Features

- Priorities from 1 (highest) to 10 (lowest)
- Exponential back-off with jitter
- Per-queue rate limits, for example 100 jobs per minute
- Pluggable storage: memory, Redis or SQLite
- Typed events for `completed`, `failed` and `stalled` jobs

Installation

Install the package from npm:

```
npm install fastqueue
```

Node.js 18 or later is required. The package ships its own type definitions.

Quick start

Create a queue, add a worker, then push jobs:

```
import { Queue } from "fastqueue";

const queue = new Queue("emails", { concurrency: 4 });

queue.process(async (job) => {
  await sendEmail(job.data.to, job.data.subject);
});

await queue.add({ to: "ana@example.com", subject: "Welcome" }, { priority: 2 });
```

Options

Option	Type	Default	Description
`concurrency`	number	`1`	Jobs processed at the same time

Option	Type	Default	Description
`retries`	number	`3`	Attempts before a job fails
`backoff`	string	`exponential`	`fixed` or `exponential`
`rateLimit`	object	none	`{ max, duration }` in milliseconds
`storage`	string	`memory`	`memory`, `redis` or `sqlite`

Roadmap

- Priorities and retries
- Redis storage
- Scheduled (delayed) jobs
- Web dashboard

Contributing

1. Fork the repository and create a branch.
2. Run the tests with `npm test`.
    - Unit tests run in under ten seconds.
    - Integration tests need a local Redis server.
3. Open a pull request that describes the change.

Please read the code of conduct before you open an issue.

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